

Mohammad Ali

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RESEARCH INTERESTS

Computer Systems and Applications, ML Systems, Scalable Machine Learning Pipelines, Multi-Task Learning, Explainable AI

EDUCATION

Ph.D (Computer Science)	2026
Indraprastha Institute of Information Technology Delhi (IIIT Delhi)	
Bachelor of Technology in Computer Engineering	2022 – 2026
Zakir Husain College of Engineering and Technology, Aligarh Muslim University	CGPA: 9.033/10
AISSCE (Class XII)	2022
St. Fidelis Sr. Sec. School, Aligarh	Score: 94%

PUBLICATIONS

F. Alam, M. Ali, N. Akhtar, “An Artificial Intelligence and Machine Learning–Based In-Silico Framework for Novel Inhibitor Discovery in Diabetes Mellitus,” *Proc. IMPACT 2026*, Aligarh, India, Feb. 2026. Available: IEEE Xplore **Best Paper Award (AI/ML Applications)**

RESEARCH EXPERIENCE

Indo-US Research Experience Undergraduate Program	Feb 2025 – Jul 2025
Ohio State University – AMU Collaboration	Aligarh, India
Supervisor: Dr. Nadeem Akhtar	

- Conducted full-cycle research leading to an **IEEE IMPACT 2026 accepted paper**.
- Designed **ML system pipelines** for QSAR-based bioactivity prediction using **large-scale chemical datasets**.
- Developed Random Forest, XGBoost, and **Multi-View Multi-Task Learning (MV-MTL)** models.
- Applied **Explainable AI (SHAP)** for feature attribution and system-level interpretability.
- Achieved **Pearson correlation up to 0.95** on independent test sets.

Summer Research Intern	Jun 2025 – Aug 2025
Centre for AI Research in Sports Analytics, IIIT Nagpur	Nagpur, India

- Developed predictive systems for **English Premier League (EPL)** match outcome forecasting.
- Engineered **feature pipelines** using player, team, and contextual statistics.
- Implemented ensemble models using **XGBoost and Logistic Regression** with SMOTE.
- Achieved **65% predictive accuracy** and analyzed system limitations for future research.

Summer Research Intern	Jul 2024 – Sep 2024
Centre of Advanced Research in Electrified Transportation (CARET), AMU	Aligarh, India

- Developed ML-based systems for **Battery Management Systems (BMS)**.
- Focused on **State of Charge (SoC)** estimation and fault analysis of Li-ion batteries.
- Achieved **95% accuracy** with significantly reduced RMSE scores.

PROJECTS

NAFLD Prediction System Machine Learning	2024
Designed a non-invasive ML system for early detection of liver disease; applied RFE and Chi-Square feature selection; utilized CTGAN for class imbalance; achieved F1-score of 0.92 .	

VitalOps | IoT & DevOps2025

Designed a **distributed real-time health monitoring system** integrating IoT sensors, FastAPI backend, **Dockerized deployment**, and **CI/CD pipelines**, focusing on system reliability and scalability.

SKILLS

Programming: Python (Advanced), C, C++ , JavaScript
Machine Learning: Random Forest, XGBoost, **Deep Learning**, Multi-Task Learning, Feature Engineering
Systems & Tools: FastAPI, **Docker**, REST APIs, **CI/CD (GitHub Actions)**, Linux Environment
Frameworks: PyTorch, TensorFlow, Scikit-Learn, SHAP, CTGAN, OpenCV
Data Engineering: SQL, MongoDB, Large-Scale Data Preprocessing
Research Tools: Git, Jupyter Notebook, Google Colab, **LaTeX**

LEADERSHIP & ACADEMIC SERVICE

Coordinator, AMURoboClub <i>ZHCET, AMU, Aligarh</i>	Jun 2025 – Present
• Led technical planning for robotics workshops and hands-on training sessions. • Mentored 50+ junior students in embedded systems and algorithmic problem-solving. • Coordinated interdisciplinary student teams for national-level project demonstrations.	
Technical Lead, Google Developer Groups Club <i>ZHCET, AMU, Aligarh</i>	Aug 2024 – May 2025
• Organized Cloud Computing and DevOps workshops for the student community. • Introduced students to Scalable Systems and industry-standard CI/CD practices.	